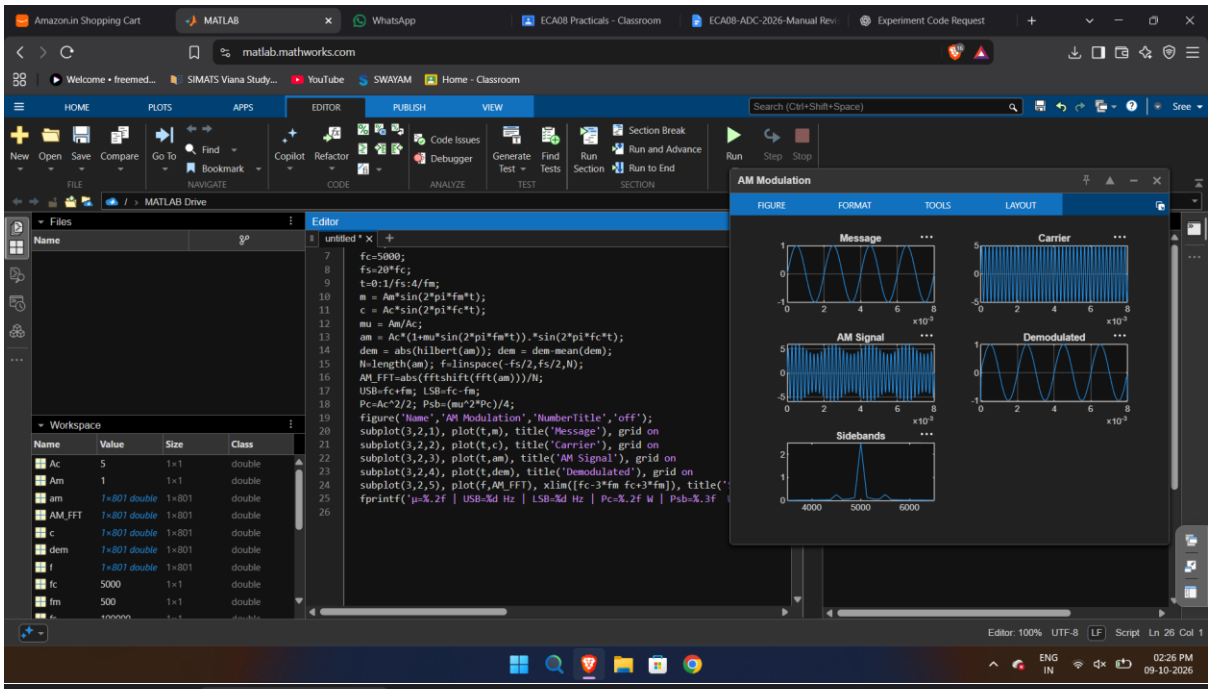
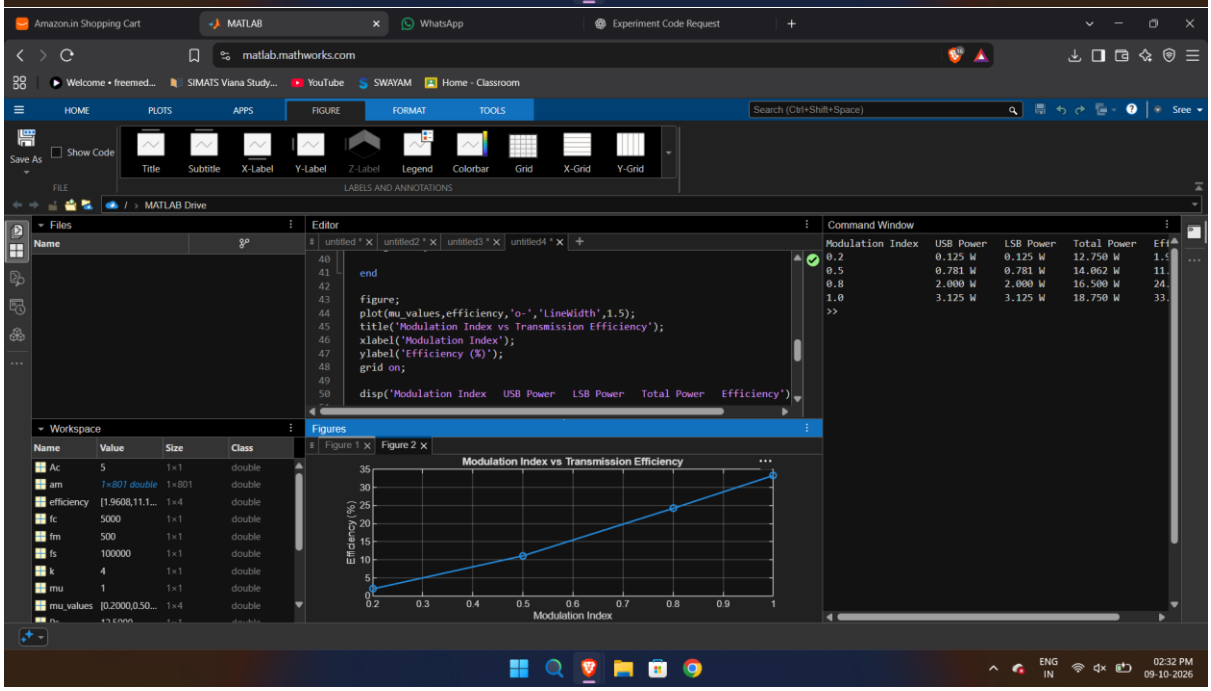
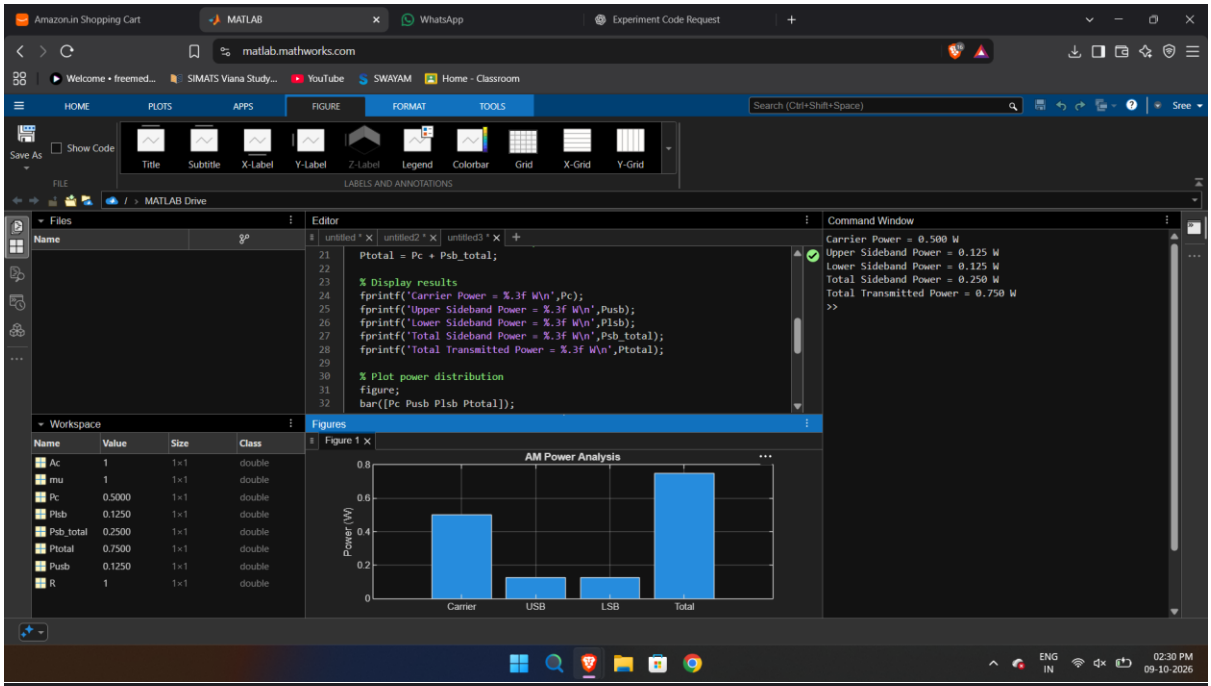
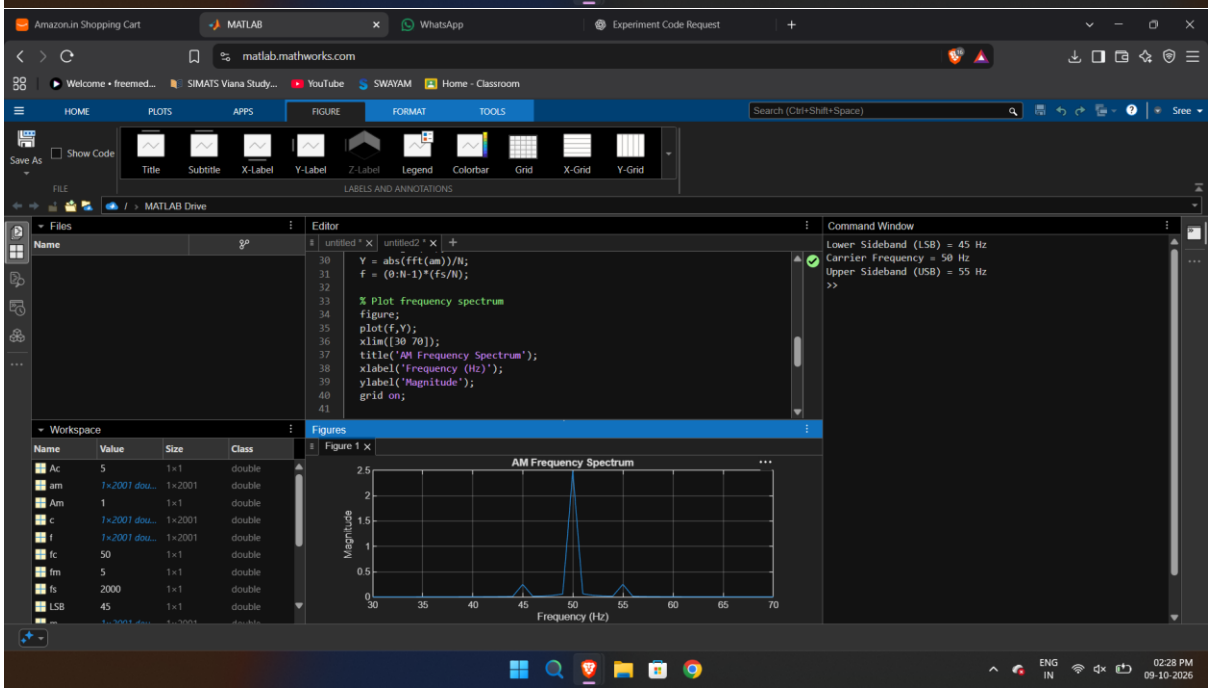
EX : 2 - CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB






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Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
Editor
untitled *
untitled2 *
untitled3 *
21 Ptotal = Pc + Psb_total;
22
23 % Display results
24 fprintf('Carrier Power = %.3f W\n',Pc);
25 fprintf('Upper Sideband Power = %.3f W\n',Pusb);
26 fprintf('Lower Sideband Power = %.3f W\n',Plsb);
27 fprintf('Total Sideband Power = %.3f W\n',Psb_total);
28 fprintf('Total Transmitted Power = %.3f W\n',Ptotal);
29
30 % Plot power distribution
31 figure;
32 bar([Pc Pusb Plsb Ptotal]);
Command Window
Carrier Power = 0.500 W
Upper Sideband Power = 0.125 W
Lower Sideband Power = 0.125 W
Total Sideband Power = 0.250 W
Total Transmitted Power = 0.750 W
>>
Workspace
Name Value Size Class
Ac 1 1×1 double
mu 1 1×1 double
Pc 0.5000 1×1 double
Plsb 0.1250 1×1 double
Psb_total 0.2500 1×1 double
Ptotal 0.7500 1×1 double
Pusb 0.1250 1×1 double
R 1 1×1 double
Figures
Figure 1
AM Power Analysis
Power (W)
0
0.2
0.4
0.6
0.8
Carrier
USB
LSB
Total
ENG
IN
02:30 PM
09-10-2026


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Colorbar
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X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
Editor
untitled *
untitled2 *
30 Y = abs(fft(am))/N;
31 f = (0:N-1)*(fs/N);
32
33 % Plot frequency spectrum
34 figure;
35 plot(f,Y);
36 xlim([30 70]);
37 title('AM Frequency Spectrum');
38 xlabel('Frequency (Hz)');
39 ylabel('Magnitude');
40 grid on;
41
Command Window
Lower Sideband (LSB) = 45 Hz
Carrier Frequency = 50 Hz
Upper Sideband (USB) = 55 Hz
>>
Workspace
Name Value Size Class
Ac 5 1×1 double
am 1×2001 dou... 1×2001 double
Am 1 1×1 double
c 1×2001 dou... 1×2001 double
f 1×2001 dou... 1×2001 double
fc 50 1×1 double
fm 5 1×1 double
fs 2000 1×1 double
LSB 45 1×1 double
Figures
Figure 1
AM Frequency Spectrum
Magnitude
0
0.5
1
1.5
2
2.5
30
35
40
45
50
55
60
65
70
Frequency (Hz)
ENG
IN
02:28 PM
09-10-2026